EML Trees Are Universal Approximators

Anonymous Author(s)
Affiliation
Address
email

Abstract

The recently introduced EML (Exp-Minus-Log) function acts as continuous analogue of NAND gates, providing a compositional building block capable of representing elementary functions. In this work, we study the expressive power of tree-structured compositions of EML functions. We show that such trees enjoy a universal approximation property for functions in $W^{k,\infty}$ for $k \in \mathbb{N}$, drawing on classical neural network approximation arguments while exploiting the ability to explicitly construct EML trees that mimic polynomial representations. We further propose a learning algorithm for EML-type trees equipped with fitting parameters, and demonstrate its feasibility in practical optimization problems. Our results establish EML trees as a theoretically grounded framework for function approximation.

1 Introduction

$$\mathrm{EML}(x, y) = e^x - \operatorname{Log} y \tag{1}$$

Analogy to NAND gates

Mention that EML is a mathematical curiosity, more than a new paradigm for function approximation

State that since we have all elementary functions, that we can recognize that we should expect an approximation theorem, since we can build polynomial

2 Background

EQL, KAN, AI-Feynman, SINDy (/ PDEFind), Symbolic regression, SymTorch, Bacon (Langley)

Symbolic metamodel:

- Demystifying Black-box Models with Symbolic Metamodels (NeurIPS)

3 Universal Approximation Theorem

For our work, we consider a generalization of the EML function (1), denoted EML_{θ_i}, to include 6 learnable parameters each, and represented as

$$\mathrm{EML}_{\theta_i}(x, y) = a_i e^{b_i x + c_i} + d_i \ln(e_i y + f_i), \tag{2}$$

where i enumerates the instance of the EML function in the tree and

$$\theta_i = \{a_i, b_i, c_i, d_i, e_i, f_i\}. \tag{3}$$

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

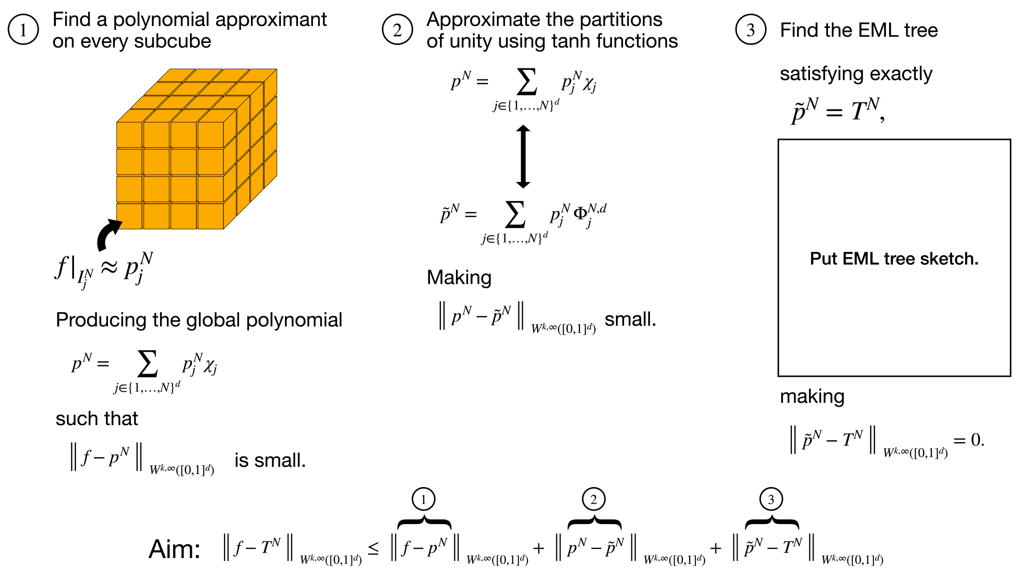


Figure 1: Sketch of the proof of the universal approximation theorem. THIS IS AN OLDER SCHEMATIC; NEEDS UPDATE.

For ..., we will often refer to individual EML_{θ_i} as “atoms" and the reusable functions or operations built using a collection of EML’s with fixed parameters as “blocks".

Of course, one observes that setting $a_i = b_i = e_i = 1$, $d_i = -1$, and $c_i = f_i = 0$ recovers the original EML function (1).

The purposes behind this generalization are both technical and practical.

As will be detailed in the upcoming subsections, proving the approximation theorem relies on exactly representing a polynomial as an EML tree, but the technical lemma 2, which allows us to approximate the original function by a polynomial does not give us an estimate or an upper bound on the coefficients of these polynomials; thus, with the current framework, it is impossible to know the size of the EML tree that represents such polynomials, because in the original function (1), constants need to build recursively from smaller ones (starting from 1), so reaching coefficients badly depends on their magnitude.

Additionally, practically, as we will witness when we construct operations using these generalized EML functions, they are able to represent operations and functions more compactly, as will be demonstrated for the basic binary operations $(+, -, \times, \div)$ and others used in building polynomials.

It also performs better.

Another advantage is that we now no longer need to consider the number 1 as a possible input to the EML atoms, thus reducing the inputs to the EML atoms as either the variables or other EML atoms.

3.1 Setting

3.2 Statement of the Theorem

Theorem 1 (Universal Approximation Theorem in $W^{k,\infty}$)**.**

3.3 Outline of the Proof

3.4 Remarks

Remark 1. *For non positive functions.*

Remark 2. *For recovering trigonometric functions.*

Remark 3. *There is an inherent nonuniqueness associated with these generalized EMLs. (comment on addition block for example).*

Remark 4. *Corollary to any domain via affine transformation.*

Remark 5. *Removing the need for approximating indicator functions by adding ReLU function to definition of EML (?).*

4 Implementation

The constructive proof of Theorem 1 establishes that the parametric EML grammar can express, exactly, any element of an arbitrarily fine polynomial approximant to a $W^{k,\infty}$ function. It does not by itself address whether such a tree can be *recovered* from data by gradient-based optimisation. In contrast to Odrzywołek [2026], which targets *symbolic* recovery of vanilla-EML trees over the discrete grammar $\{1, x, \mathrm{EML}(\cdot,\cdot)\}$ and reports rapid collapse of recovery rate at depth ≥ 5, we exploit the continuous parameters of the generalised atom $\mathrm{EML}_{\theta_i}(x, y)$ to make the optimisation landscape amenable to first-order methods. This section reports a small, focused empirical study in support of the theorem's practical content.

4.1 Setup

Architecture. We use the PEM tree of Section 1 directly: a complete binary tree T_θ of $2^d - 1$ atoms, each carrying six free parameters $\theta_i = \{a_i, b_i, c_i, d_i, e_i, f_i\}$ and computing $\mathrm{EML}_{\theta_i}(x, y)$. Children are fed *raw* between atoms, with the input variable x entering at the leaf-parents directly. For univariate x the parameter count is $\#\theta(d) = 6(2^d - 1) \in \{42, 90, 186\}$ at depths $d \in \{3, 4, 5\}$.

Two parameterisations. We consider two regimes:

Real (R). All $\theta_i \in \mathbb{R}^6$. To keep the operator real on $\mathbb{R}$, the logarithm's argument is routed through $\ln(\mathrm{softplus}(\cdot) + \epsilon)$ with $\epsilon = 10^{-6}$, a smoothed-positive surrogate for the principal-branch logarithm. This blocks the trigonometric path of the corresponding remark in Section 1 but eliminates branch-cut pathologies.

Complex (C). All $\theta_i \in \mathbb{C}^6$, stored as twin real Parameters (θ_i^R, θ_i^I). Forward in $\mathbb{C}^{128}$ with the principal branch of $\ln$; the loss is computed on $\mathrm{Re}(T_\theta(x))$. We add a small auxiliary penalty $\lambda \cdot \overline{|\mathrm{Im}\, T_\theta|^2}$ with $\lambda = 10^{-3}$ so that the trained predictor is honestly real-valued. Doubling the parameter count is the cost; gaining the trigonometric path is the intended benefit.

Optimisation procedure. We minimise mean-squared error on a uniform grid of $N = 200$ samples per target with the following two-stage strategy, applied per (target, depth, parameterisation) cell at a fixed wall-clock budget of 30 seconds:

1. **Adam multi-restart.** From independent Gaussian-initialised seeds, run 1500 steps of Adam with learning rate 5×10^{-3}, cosine-annealed to zero, gradient ℓ_2-clip at 5.0. Track the best mean-squared error across restarts.
2. **L-BFGS refinement.** Apply L-BFGS with strong-Wolfe line search (≤ 200 iterations, history 20) to the best-of-restarts model and take the smaller of the pre- and post-refinement losses.

The number of Adam restarts per cell is determined by the budget rather than fixed and is reported in Table 1.

Assumptions. The reported numbers depend on three explicit choices. (i) Under R the operator is the softplus-stabilised log surrogate, not vanilla EML; the two coincide on positive arguments and diverge smoothly near zero. (ii) All targets are noise-free, dense samples on the stated domain; no claim is made about generalisation to sparse or noisy data. (iii) The wall-clock budget is small (30 s on a single CPU) and held fixed across cells; reported errors at the larger depths are optimisation-limited, not capacity-limited.

4.2 Targets and metric

We report results on five 1D targets, each chosen to align with a specific construction in the proof:

Table 1: Best relative RMSE per cell, PEM trees with real (R) and complex (C) parameters at depths $3, 4, 5$. Strategy: Adam multi-restart + L-BFGS, 30 s wall-clock per cell. Parenthesised integer is the number of completed Adam restarts within the budget; parameter count is $\#\theta = 6\,(2^d - 1)$ for R and twice that for C. Bolded entry per row is the best of the six.

Target	**Real (R)**, $\#\theta = 42, 90, 186$			**Complex (C)**, $\#\theta = 84, 180, 372$		
	d=3	d=4	d=5	d=3	d=4	d=5
$x^3 - x$	$7.58{\cdot}10^{-3}$ (15)	$\mathbf{1.47{\cdot}10^{-3}}$ (8)	$4.32{\cdot}10^{-3}$ (4)	$2.17{\cdot}10^{-2}$ (7)	$1.37{\cdot}10^{-2}$ (4)	$5.20{\cdot}10^{-2}$ (2)
$\tanh(2x)$	$4.20{\cdot}10^{-3}$ (15)	$\mathbf{1.22{\cdot}10^{-3}}$ (8)	$2.10{\cdot}10^{-3}$ (4)	$1.77{\cdot}10^{-3}$ (7)	$5.16{\cdot}10^{-3}$ (4)	$1.80{\cdot}10^{-2}$ (2)
$\sin(x)$	$\mathbf{3.02{\cdot}10^{-3}}$ (15)	$7.67{\cdot}10^{-3}$ (8)	$1.01{\cdot}10^{-2}$ (4)	$1.74{\cdot}10^{-2}$ (7)	$3.49{\cdot}10^{-3}$ (4)	$2.07{\cdot}10^{-2}$ (2)
e^{-x^2}	$4.13{\cdot}10^{-3}$ (15)	$\mathbf{1.12{\cdot}10^{-3}}$ (7)	$4.66{\cdot}10^{-3}$ (4)	$2.20{\cdot}10^{-2}$ (7)	$5.27{\cdot}10^{-3}$ (3)	$1.27{\cdot}10^{-2}$ (2)
$\sin(3x)\,e^{-x^2/2}$	$\mathbf{3.23{\cdot}10^{-2}}$ (15)	$9.80{\cdot}10^{-2}$ (7)	$9.09{\cdot}10^{-2}$ (4)	$6.45{\cdot}10^{-2}$ (6)	$3.91{\cdot}10^{-2}$ (3)	$3.57{\cdot}10^{-2}$ (2)

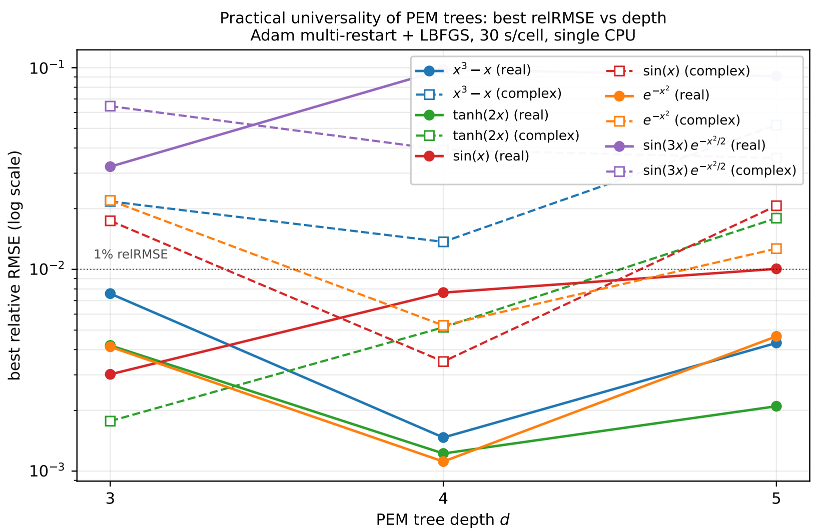


Figure 2: Best relative RMSE vs PEM tree depth, five targets, real (solid) and complex (dashed) parameterisations, 30 s wall-clock per cell. The dotted line marks the 1% relRMSE threshold. Real PEM reaches sub-percent error on $4/5$ targets by depth 4; complex PEM is competitive only on $\sin(3x)\,e^{-x^2/2}$, where its larger capacity helps despite the halved restart count.

- $x^3 - x$ on $[-1.5, 1.5]$: a degree-3 polynomial; the local approximant of Lemma 2 in its simplest non-trivial form. The constructive bound is $5k - 3 = 12$ atoms for degree $k = 3$.
- $\tanh(2x)$ on $[-2, 2]$: hyperbolic tangent, used in the partition-of-unity construction (Section 6); the explicit EML realisation is in Figure 5.
- $\sin(x)$ on $[-\pi, \pi]$: the trigonometric case highlighted in the corresponding remark; the vanilla-EML construction routes through $\ln(-1) = i\pi$, unreachable from R.
- e^{-x^2} on $[-2.5, 2.5]$: smooth analytic, no oscillation — baseline for "easy" targets.
- $\sin(3x)\,e^{-x^2/2}$ on $[-3, 3]$: oscillation $\times$ envelope; the hardest target in this set.

We report *relative RMSE*, $\mathrm{relRMSE} = \|T_\theta(x) - f(x)\|_2 / \|f(x)\|_2$, on the training grid, best-of-restarts.

4.3 Results

Table 1 reports the best relRMSE per cell. Figure 2 plots relRMSE against depth on a log scale.

What the numbers say. Three observations support the theorem's practical content. **(i)** Real-parameterised PEM trees achieve sub-percent relRMSE on 4 of 5 targets at depth 4 within 30 s of single-CPU compute, including the polynomial target $x^3 - x$ which falls to 1.47×10^{-3}. This is a direct empirical reflection of the constructive route of the theorem (polynomial approximation, then exact EML representation): gradient methods reach close to the constructed optimum, even though the construction itself is not used as an initialisation. **(ii)** The trigonometric target $\sin(x)$ does *not* require the complex parameterisation in our setting: real PEM at depth 3 already reaches 3.0×10^{-3}. The softplus-stabilised log is expressive enough on this domain that the complex path is not the binding constraint at the budget shown. **(iii)** The hardest target, $\sin(3x)\, e^{-x^2/2}$, is the one clear case where the complex parameterisation pays off: 3.6×10^{-2} at depth 5 (C) versus 9.1×10^{-2} at the same depth (R). The composite oscillation-envelope target rewards the extra capacity even though C completes only half as many restarts in the same wall-clock budget.

The general pattern visible in Figure 2 — error falling sharply from depth 3 to depth 4, then rising at depth 5 for several real targets — is a wall-clock-budget effect: deeper trees have more parameters but, at fixed budget, fewer restarts to find good basins. With unlimited compute this artefact would disappear; under fixed compute it suggests an "elbow" depth around $d = 4$ for these 1D targets.

Computational cost. The full sweep (5 targets × 3 depths × 2 parameterisations = 30 cells) completes in approximately 12 minutes on a single laptop CPU (Apple M-series, no GPU). Per-cell wall-clock is 30 s by construction; the number of completed Adam restarts ranges from ~ 15 at $(d{=}3, \mathrm{R})$ down to 2 at $(d{=}5, \mathrm{C})$. No GPU acceleration or parallelism was used.

4.4 Limitations

We list the limitations of this empirical study honestly, in priority order.

- **Single-run-of-experiment.** Tabled values are the best loss from one run of the multi-restart procedure. Variance across full reruns is not reported. Order-of-magnitude conclusions are robust; comparisons within a factor of ~ 2 should not be over-interpreted.
- **Real-softplus surrogate.** The R parameterisation is not vanilla EML. Trees fit under R coincide with EML on the positive-argument slice but smoothly diverge near zero. The complex parameterisation uses the operator literally.
- **Univariate, smooth, dense, noise-free.** Multivariate fitting, sparse sampling, and noise robustness are all open and likely harder; the theorem is dimension-agnostic but our experiments are not.
- **Fixed-budget effects on depth comparison.** Deeper trees receive fewer restarts at fixed wall-clock, so the depth-vs-error curve confounds capacity gain with restart-count loss. The dip at depth 4 for several targets is the visible signature.
- **Optimisation, not approximation, theory.** The reported numbers are achieved errors, not bounds. They are upper-bounded by the theorem; whether they meet the rate the theorem predicts (and at what budget the gap closes) is a question we do not address here.

5 Discussion

- Generalization to other spaces (Besov)
- Remark on analytic functions
- Finding a way to use the vanilla EMLs by following another proof technique that involves finding the best polynomial approximant with bounded coefficients and finding upper bounds for constructing these constants (which is a challenging task to get the most optimal upper bounds for getting high constants).
- We could maybe make due with less than 6 parameters per EML atom.
- Trying to find better techniques to optimize these trees.
- Probably shorter trees exist that do the same thing; but, in this case, they were constructed like this to be human readable and easy to verify and interpretable.
- Can we find another function that trades depth for "width", by resorting to making the atoms take in more inputs (similar to network constructions)
- Practically, the logarithm takes in only positive values, so we could put an absolute value inside the logarithm to circumvent any potential issue with domain

- Discussion when talking about possible use for interpretable scientific discovery (Wyder et al. [2025], Yermakov et al. [2025])

Open Question 1. *Can we*

A Notation

$\mathbb{N}^d$

$\times$ denotes cartesian product

B Sobolev Spaces

We recall the definitions of some basic functional spaces.

The Lebesgue spaces, denoted $L^p(\Omega)$ are defined as

$$L^p(\Omega) = \left\{ f : \Omega \to \mathbb{R} \;\middle|\; \int_\Omega |f(x)|^p \, dx < \infty \right\}, \tag{4}$$

for $1 \leq p < \infty$, equipped with the norm

$$\|f\|_{L^p(\Omega)} = \left(\int_\Omega |f(x)|^p \, dx \right)^{\frac{1}{p}}. \tag{5}$$

When $p = \infty$, where $L^\infty(\Omega)$ consists of functions that are essentially bounded and whose norm is

$$\|f\|_{L^\infty(\Omega)} = \operatorname*{ess\,sup}_{x \in \Omega} |f(x)| = \inf \{ C \geq 0 \mid |f(x)| \leq C \text{ for almost every } x \in \Omega \}. \tag{6}$$

Moreover, the Sobolev space $W^{k,p}$ is defined as

$$W^{k,p}(\Omega) = \left\{ f \in L^p(\Omega) \mid D^\alpha f \in L^p(\Omega) \text{ for all } \alpha \in \mathbb{N}_0^d \text{ with } |\alpha| \leq k \right\}. \tag{7}$$

We also define associated seminorms. For $p < \infty$, we have

$$|f|_{W^{m,p}(\Omega)} = \left(\sum_{|\alpha| = m} \|D^\alpha f\|_{L^p(\Omega)}^p \right)^{\frac{1}{p}}, \qquad \text{for } m = 0, \ldots, k, \tag{8}$$

and for $p = \infty$, we have

$$|f|_{W^{m,\infty}(\Omega)} = \max_{|\alpha| = m} \|D^\alpha f\|_{L^\infty(\Omega)}, \qquad \text{for } m = 0, \ldots, k. \tag{9}$$

Based on these seminorms, we can define the following norms for $p < \infty$,

$$\|f\|_{W^{k,p}(\Omega)} = \left(\sum_{m=0}^{k} |f|_{W^{m,p}(\Omega)}^p \right)^{\frac{1}{p}}, \tag{10}$$

and for $p = \infty$,

$$\|f\|_{W^{k,\infty}(\Omega)} = \max_{0 \leq m \leq k} |f|_{W^{m,\infty}(\Omega)}. \tag{11}$$

In particular, when $p = 2$, the Sobolev spaces $H^k(\Omega) = W^{k,2}(\Omega)$ (with $k \in \mathbb{N}_0$) are Hilbert spaces with the corresponding norms $\|\cdot\|_{H^k(\Omega)} = \|\cdot\|_{W^{k,2}(\Omega)}$ and seminorms $|\cdot|_{H^m} = |\cdot|_{W^{m,2}}$ for any integer m such that $0 \leq m \leq k$.

Additionally, the space of k-times continuously differentiable functions $C^k(\Omega)$, with the associated norm $\|f\|_{C^k(\Omega)} = \|f\|_{W^{k,\infty}(\Omega)}$.

We will make frequent use of the following Banach algebra property.

Lemma 1. *Let $d \in \mathbb{N}$, $k \in \mathbb{N}_0$, and $\Omega \subset \mathbb{R}^d$. If $f, g \in W^{k,\infty}(\Omega)$, then*

$$\|fg\|_{W^{k,\infty}(\Omega)} \leq 2^k \|f\|_{W^{k,\infty}(\Omega)} \|g\|_{W^{k,\infty}(\Omega)}. \tag{12}$$

C Useful Definitions and Lemmas

We recall the definition of a star-shaped set, which is a necessary geometric requirement for the approximation of functions by polynomials in Sobolev spaces (in Lemma 2).

Definition 1. *A set $A \subset \mathbb{R}^d$ is star-shaped with respect to every point in a subset $B \subset A$ if*

$$\forall y \in B, \forall x \in A, [y, x] \subset A, \tag{13}$$

where $[y, x] = \{(1-t)y + tx \mid t \in [0,1]\}$ is the straight line in $\mathbb{R}^d$ connecting the point y to the point x.

We recall the Bramble-Hilbert lemma for Sobolev spaces, as proved in De Ryck et al. [2021], with similar statements appearing in Durán [1983] and Verfürth [1999].

Lemma 2. *Let $\Omega \subset \mathbb{R}^d$ be an open and bounded set of diameter $0 < h < e^{-1/2}d^{-3/2}$ which is star-shaped with respect to every point in an open ball $B \subset \Omega$ with diameter ρh. Then, for every $f \in W^{s,\infty}(\Omega)$, there exists a polynomial $\hat{f}$ of degree at most $s-1$ such that for any $k \in \mathbb{N}_0$ with $k < s$, it holds that*

$$\|f - \hat{f}\|_{W^{k,\infty}(\Omega)} \le \frac{\pi^{1/4}\sqrt{s}(d\sqrt{de}h)^{s-k}}{(s-k-1)!} |f|_{W^{s,\infty}(\Omega)}. \tag{14}$$

D Required Building Blocks

In this section, we present the explicit constructions of the basic binary operations, exponentiation, and the hyperbolic tangent function, all of which are used in the approximation proof (Theorem 1).

D.1 Basic Binary Operations

In Figure 3, we provide the EML trees (with their associated parameters) to produce the basic binary operations of addition, subtraction, mutiplication, and division. The addition and subtraction blocks are used in the constructions of the multiplication and division blocks respectively.

As observed in Figure 3, we note that the size and depth requirements of these binary operations are summarized in Table 2.

Operator	$N(\cdot)$	$\mathrm{Depth}(\cdot)$
$x +_\theta y$	3	2
$x -_\theta y$	3	2
$x \times_\theta y$	6	4
$x \div_\theta y$	6	4
$x_\theta^n,\ mx_\theta^n$	2	2
$\tanh_\theta(x),\ m\tanh_\theta(x)$	8	5

Table 2: Size and depth of the operations in Appendix D.

D.2 Exponentiation

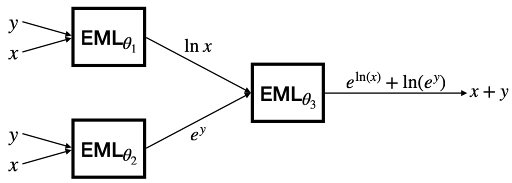
In our proof, we will also make frequent use of exponentiation to a constant power. We provide explicit constructions for EML trees that perform x^n and mx^n for constants $m, n \in \mathbb{R}$ in Figure 4.

Note that in Figure 4b, we construct mx^n directly, which we use to represent the monomial terms of polynomials (in Appendix E). This is because it is more efficient to incorporate the coefficient within the exponentiation block itself, rather than introducing a separate multiplication block to combine a constant with the x^n block.

For both exponentiation blocks (4a and 4b), the size and depth of the associated EML trees are 2 (and appended to Table 2).

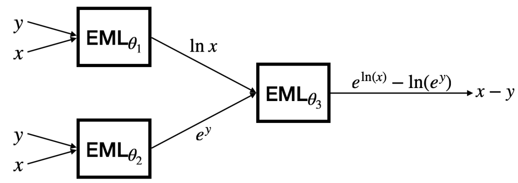
D.3 Hyperbolic Tangent

The hyperbolic tangent is constructed, using the exponential definition, by first getting $e^{2x} - 1$ and $e^{2x} + 1$ (via two separate EML blocks) and then diving them, as in Figure 5. The size of the $\tanh$ block is 8 and its depth is 5, which are appended to Table 2.



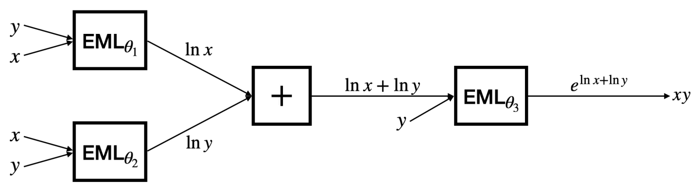
$\theta_1 = \{0,0,0,1,1,0\}$,
with $\theta_2 = \{1,1,0,0,0,1\}$,
$\theta_3 = \{1,1,0,1,1,0\}$.

(a) Addition (3)



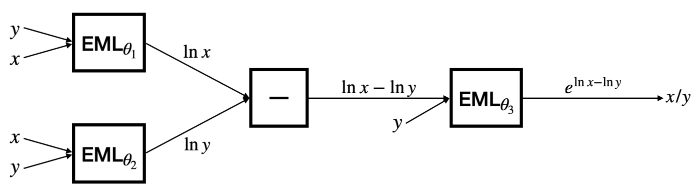
$\theta_1 = \{0,0,0,1,1,0\}$,
with $\theta_2 = \{1,1,0,0,0,1\}$,
$\theta_3 = \{1,1,0,\text{-}1,1,0\}$.

(b) Subtraction (3)



$\theta_1 = \{0,0,0,1,1,0\}$,
with $\theta_2 = \{0,0,0,1,1,0\}$,
$\theta_3 = \{1,1,0,0,0,1\}$.

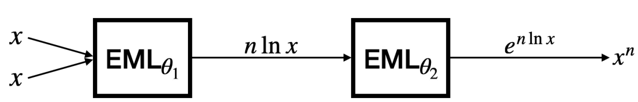
(c) Multiplication (6)



$\theta_1 = \{0,0,0,1,1,0\}$,
with $\theta_2 = \{0,0,0,1,1,0\}$,
$\theta_3 = \{1,1,0,0,0,1\}$.

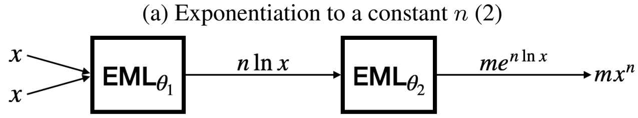
(d) Division (6)

Figure 3: The Basic Binary Operations



with $\theta_1 = \{0,0,0,n,1,0\}$,
$\theta_2 = \{1,1,0,0,0,1\}$.

(a) Exponentiation to a constant n (2)



with $\theta_1 = \{0,0,0,n,1,0\}$,
$\theta_2 = \{m,1,0,0,0,1\}$.

(b) Exponentiation to a constant n with prefactor m (2)

Figure 4: Exponentiating

We can similarly obtain $m \tanh x$, which is useful for the construction of the approximate partitions of unity, by amending the division block to include the prefactor m. This is done by changing θ_3 in the division block (Figure 3d) to be $\theta_3 = \{m, 1, 0, 0, 0, 1\}$.

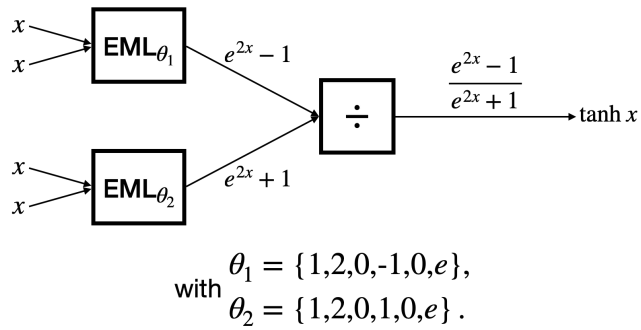


Figure 5: Hyperbolic Tangent (8)

E Constructing Polynomials

The following two subsections detail how to construct univariate and multivariate polynomials using EML trees and provide upper bounds for the number of required EML blocks in each case.

E.1 Univariate Polynomials

A univariate polynomial of degree k, denoted by

$$p(x) = \sum_{i=0}^{k} a_i x^i = a_0 + a_1 x + a_2 x^2 + \cdots + a_k x^k, \tag{15}$$

can be (naively) constructed using EML blocks as in Figure 6, wherein we used two simple constituent blocks: addition (Figure 3a) and exponentiation to a constant with a prefactor (Figure 4b).

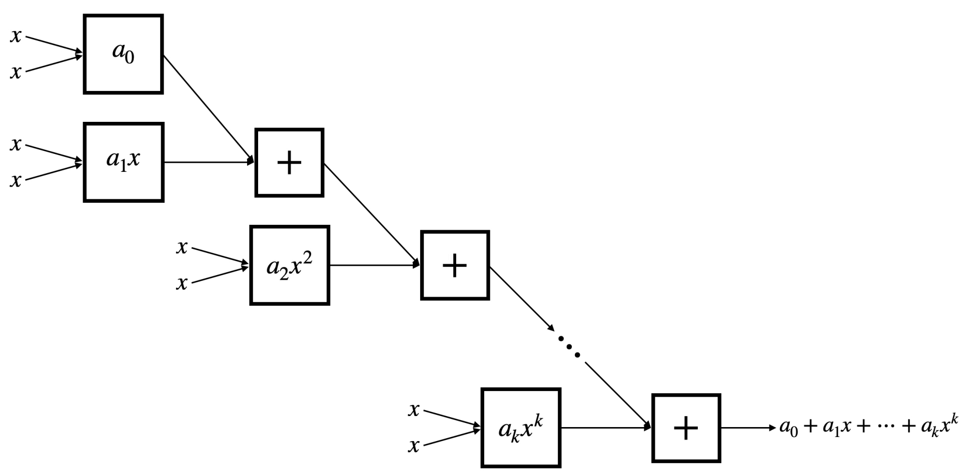


Figure 6: Univariate Polynomial $(5s - 3)$

The upper bound on the number of required EML blocks is obtained by noting that we used

- $k - 1$ addition blocks (each requiring 3 EML atoms), and
- k exponentiation with a prefactor blocks (each requiring 2 EML atoms).

So, constructing an EML tree p_θ such that $p_\theta = p$ (as per the above procedure) requires $N(T_\theta) = 3(k-1) + 2k = 5k - 3$ and $\text{Depth}(T_\theta) = 2 + 2(k-1) = 2k$, since the longest path starts with the x feeding into the a_0 block and passes through all $k - 1$ addition blocks.

Hence, we phrase the above conclusion as the lemma:

Lemma 3 (Construction of Univariate Polynomials)**.** *Given a univariate polynomial p of degree k, there exists an EML tree p_θ (following the explicit construction in Figure 6), such that $p_\theta = p$ with $N(p_\theta) \le 5k - 3$ and Depth$(p_\theta) \le 2k$.*

E.2 Multivariate Polynomials

We recall that, for $\mathbf{x} = (x_1, \dots, x_d) \in \mathbb{R}^d$, a multivariate polynomial p of degree k is of the form

$$p(\mathbf{x}) = \sum_{|\alpha| \le k} c_\alpha x_1^{\alpha_1} x_2^{\alpha_2} \cdots x_d^{\alpha_d}, \tag{16}$$

where $\alpha = (\alpha_1, \alpha_2, \dots, \alpha_d)$ is a multi-index with $|\alpha| = \alpha_1 + \alpha_2 + \cdots + \alpha_d$, and c_α are the associated coefficients. Such a polynomial (of degree $\le k$) has $\binom{d+k}{k}$ monomial terms.

To construct an EML tree for the polynomial (16), (1) we construct the monomials $c_\alpha x_1^{\alpha_1}$, $x_2^{\alpha_2}$, $\dots$, $x_d^{\alpha_d}$ for each α with $|\alpha| \le k$, (2) multiply these monomials together (separately for each α), (3) do steps (1) and (2) for every monomial, and (4) add them together. A breakdown of the required EML tree sizes and depths for each step is as follows:

(1) For a certain α, each monomial, of which there are d, requires an exponentiation block (and the first has a prefactor), totaling $2d$ EML atoms;

(2) Again for each α, we need to multiply the d monomials, requiring $d-1$ multiplication blocks and, thus, $6(d-1)$ EML atoms; thus, each term, considering both getting monomials and multiplying them requires $2d + 6(d-1) = 8d - 6$ EML atoms;

(3) Since we have $\binom{d+k}{d}$ terms in the polynomial, we do steps (1) and (2) $\binom{d+k}{d}$ times, thus requiring $(8d-6)\binom{d+k}{d}$ EML atoms;

(4) We finally sum over all terms, requiring $\binom{d+k}{d} - 1$ addition blocks, each consisting of 3 EML atoms.

Thus, in total, constructing the polynomial p in (16) takes

$$(8d-6)\binom{d+k}{d} + 3\left(\binom{d+k}{d} - 1\right) = (8d-3)\binom{d+k}{d} - 3 \tag{17}$$

EML atoms, and whose depth can be calculated (by drawing a schematic like Figure 6) to be

$$2 + 4(d-1) + 2\left(\binom{d+k}{d} - 1\right) = 4d + 2\binom{d+k}{d} - 4, \tag{18}$$

since the x's feed into the first $c_\alpha x^{\alpha_1}$ block (of depth 2), pass through $d-1$ multiplication (each of depth 4), and then through $\binom{d+k}{d} - 1$ addition blocks (each of depth 2).

The above justified the following lemma:

Lemma 4 (Construction of Multivariate Polynomials)**.** *Given $p : \mathbb{R}^d \to \mathbb{R}$, a multivariate polynomial of degree k, there exists an EML tree p_θ, such that $p_\theta = p$ with*

$$N(p_\theta) \le (8d-3)\binom{d+k}{d} - 3, \quad \textit{and} \quad \textit{Depth}(p_\theta) \le 4d + 2\binom{d+k}{d} - 4. \tag{19}$$

F Approximating the partition of unity

In this section, we aim to construct an approximate partition of unity, similar to Section 4 of De Ryck et al. [2021]. We do this because indicator functions are unsuitable to be represented by EML trees due to the fact that they contain a discontinuity and cannot be suitably represented by chaining elementary functions constructed from compositions of EML. For this reason, we construct smooth approximations of indicator functions on subsets of $[0,1]^d$ via $\tanh$ functions. When suitably shifted and parametrized, these functions output values close to 1 within the subset and close to 0 outside of it.

F.1 Properties of the approximate partition of unity

We first recall some of the properties of the partition of unity of De Ryck et al. [2021].

For a point $y \in \mathbb{R}$ and a parameter $\alpha \in \mathbb{R}$ (to be chosen later), we define the one-dimensional approximate indicator functions as

$$\begin{aligned}
\rho_1^N(y) &= \frac{1}{2} - \frac{1}{2}\sigma\left(\alpha\left(y - \frac{1}{N}\right)\right), \\
\rho_m^N(y) &= \frac{1}{2}\sigma\left(\alpha\left(y - \frac{m-1}{N}\right)\right) - \frac{1}{2}\sigma\left(\alpha\left(y - \frac{m}{N}\right)\right) \quad \text{for } 2 \le m \le N-1, \\
\rho_N^N(y) &= \frac{1}{2}\sigma\left(\alpha\left(y - \frac{N-1}{N}\right)\right) + \frac{1}{2}.
\end{aligned} \tag{20}$$

We can clearly see that if α is large enough, these act as soft indicator functions over the intervals $\left(\frac{m-1}{N}, \frac{m}{N}\right)$ (depending on the subscript of ρ).

To choose α, we note that since $\sigma(x) \to 1$ as $x \to \infty$ and its derivatives also saturate ($\sigma^{(m)}(x) \to 0$ with $x \to \infty$), we can choose $R > 0$ large enough such that $|\sigma^{(m)}|$ is decreasing over $[R, \infty)$ for all $1 \le m \le k$.

So, given $\varepsilon > 0$, we choose α, depending on N and ε, large enough such that

$$\frac{\alpha}{N} \ge R, \qquad 1 - \sigma\left(\frac{\alpha}{N}\right) \le \varepsilon, \qquad \alpha^m \left|\sigma^{(m)}\left(\frac{\alpha}{N}\right)\right| \le \varepsilon \text{ for all } 1 \le m \le k, \tag{21}$$

justified also by Lemma A.4 in De Ryck et al. [2021]. Consequently, α can be chosen to satisfy the above conditions and has the explicit value

$$\alpha = N \max\left\{R, \ln\left(\frac{(2k)^{k+1}(Nk)^k}{e^k \varepsilon}\right)\right\}, \tag{22}$$

(see Lemma A.5 in De Ryck et al. [2021] for justification).

Importantly, we define

$$\Phi_j^N(x) = \prod_{i=1}^{d} \rho_{j_i}^N(x_i), \tag{23}$$

which acts as our approximate partition of unity. We also define the set $\mathcal{V}_d$ as

$$\mathcal{V}_d = \left\{v \in \mathbb{Z}^d \;\middle|\; \max_{1 \le i \le d} |v_i| \le 1\right\}, \tag{24}$$

which allow us choose only the cubes directly surrounding a cube of interest.

Lemmas 5 and 6 below prove that the set of functions $\{\Phi_j^N\}_{j \in \mathcal{A}^N}$ acts as an approximate partition of unity in the sense that they satisfy the following two properties: for every $j \in \mathcal{A}^N$,

$$\sum_{v \in \mathcal{V}_d} \Phi_{j+v}^{N,d} \approx 1 \qquad \text{and} \qquad \sum_{\substack{v \notin \mathcal{V}_d, \\ j+v \in \mathcal{A}^N}} \Phi_{j+v}^{N,d} \approx 0. \tag{25}$$

We recall the below two lemmas from De Ryck et al. [2021]. The first provides a quantitative bound for how close to 1 is the contribution of Φ_j^N and its direct neighbors on I_j^N.

Lemma 5 (Lemma 4.1 of De Ryck et al. [2021])**.** *If* $0 < \varepsilon < 1/4$*, then*

$$\left\| \sum_{v \in \mathcal{V}_d} \Phi_{j+v}^N - 1 \right\|_{W^{k,\infty}(I_j^N)} \le 2^{dk} d\varepsilon. \tag{26}$$

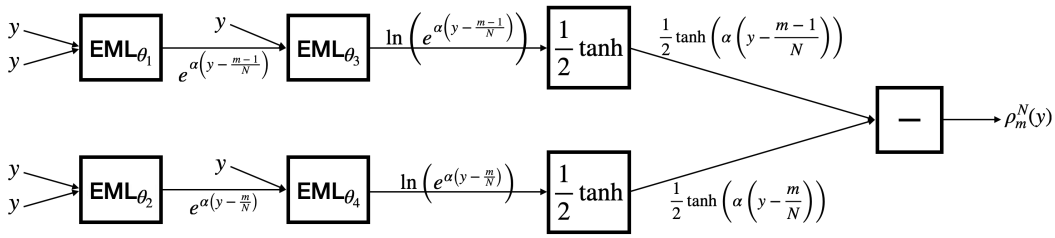
The second lemma guarantees that the contribution of the functions far away from the cube I_j^N is small.

Lemma 6 (Lemma 4.2 of De Ryck et al. [2021])**.** *Let* $k \in \mathbb{N}_0$ *and* $v \in \mathbb{Z}^d$ *with* $v_i \ge 2$ *for all* $1 \le i \le N$. *Then, it holds that*

$$\left\|\Phi_{j+v}^N\right\|_{W^{k,\infty}(I_j^N)} \le \max\{1, (2k)^{2k}\alpha^k\}\, \varepsilon. \tag{27}$$

F.2 Constructing the approximate partition of unity using EML trees

In Figure 7, we construct the ρ_m^N functions for $2 \le m \le N-1$, according to the definition in (20), and provide the associated parameters.



$$\begin{aligned} &\theta_1 = \{1, \alpha, -\alpha\frac{m-1}{N}, 0, 0, 1\}, \\ \text{with } &\theta_2 = \{1, \alpha, -\alpha\frac{m}{N}, 0, 0, 1\}, \\ &\theta_3 = \{0, 0, 0, 1, 1, 0\}, \\ &\theta_4 = \{0, 0, 0, 1, 1, 0\}. \end{aligned}$$

Figure 7: Construction of ρ_m^N for $2 \le m \le N-1$

This construction requires 4 EML blocks, 2 $\tanh$ blocks with $1/2$ constants, and a subtraction block, which boils down to a size of $4 + 2 \times 8 + 3 = 23$ and a depth of $1 + 1 + 5 + 2 = 9$.

ρ_1^N and ρ_N^N (from (20)) can be constructed more simply, requiring 3 EML blocks (one of which to get the constant $1/2$), one $\tanh$ block, and one subtraction block, thus corresponding to a size of $3 + 8 + 3 = 14$ and depth of $1 + 1 + 5 + 2 = 9$. But, for the sake of simplicity, we will consider all instances of ρ_m^N used to be those corresponding to Figure 7.

Then, we construct $\Phi_j^{N,d}$, using (23) by multiplying d instances of the one-dimensional partitions, $\rho_{j_i}^N$. This requires $d-1$ multiplication blocks, and thus the EML tree corresponding to $\Phi_j^{N,d}$ has size $23d + 6(d-1) = 29d - 6$ and depth $9 + 4(d-1) = 4d + 5$, yielding the following lemma:

Lemma 7 (Construction of Approximate Partitions of Unity)**.** *For $N, d \in \mathbb{N}$ and $j \in \mathcal{A}^N$, there exists an EML tree T_θ such that $T_\theta = \Phi_j^N$ with*

$$N(T_\theta) \le 29d - 6, \quad \text{and} \quad \mathit{Depth}(T_\theta) \le 4d + 5. \tag{28}$$

G Proof of Theorem 1

Proof. We set out to perform the steps for the proof as described in the main text above and in Figure 1.

Step 1. Approximation by Local Polynomials. We divide the domain $[0,1]^d$ into cubes of side length $1/N$. To denote these cubes, we use the corresponding index set

$$\mathcal{A}^N = \{1, \ldots, N\}^d \tag{29}$$

of cardinality $|A^N| = N^d$. For every $j \in \mathcal{A}^N$, we associate the cube defined as

$$I_j^N = \bigtimes_{i=1}^d \left(\frac{j_i - 1}{N}, \frac{j_i}{N}\right). \tag{30}$$

We also define overlapping cubes, which will be used to satisfy the condition for the Bramble-Hilbert lemma, for the given N and every $j \in \mathcal{A}^N$ as

$$J_j^N = \bigtimes_{i=1}^d \left(\frac{j_i - 2}{N}, \frac{j_i + 1}{N}\right), \tag{31}$$

which is of diameter $\frac{3\sqrt{d}}{N}$. We notice that there exists a ball of diameter $3/N$ such that J_j^N is star-shaped with respect to every point in this ball (per Definition 1). Thus, for every $j \in \mathcal{A}^N$, by the Bramble-Hilbert lemma (Lemma 2), there exists a polynomial p_j^N of degree at most $s-1$ such that, for all $0 \le k \le s-1$

$$\begin{aligned}\|f - p_j^N\|_{W^{k,\infty}(J_j^N)} &\le \frac{\pi^{1/4}\sqrt{s}}{(s-k-1)!}\left(\frac{5d^2}{N}\right)^{s-k} |f|_{W^{s,\infty}([0,1]^d)} \\ &\le \max_{0\le m\le k} \frac{\pi^{1/4}\sqrt{s}(5d^2)^{s-m}}{(s-m-1)!}\frac{|f|_{W^{s,\infty}([0,1]^d)}}{N^{s-k}} := \frac{\mathcal{C}(d,k,s,f)}{N^{s-k}},\end{aligned} \tag{32}$$

choosing $N > 5d^2$ (to obtain decay) and using $3\sqrt{e} \le 5$.

Remark 6. *The polynomials are chosen to be good approximators on the overlapping cubes (not on the disjoint ones) because*

Step 2. Definition of the Global Polynomial. We then define the global piecewise polynomial over the entire domain $[0,1]^d$ by

$$p^N = \sum_{j\in\mathcal{A}^N} p_j^N \chi_j, \tag{33}$$

where χ_j denotes the indicator function on I_j^N. Since p^N is composed using indicator functions (which are discontinuous), it cannot be adequately controlled in $W^{k,\infty}$ spaces where $k \ge 1$. This is why we introduce

$$\tilde{p}^N = \sum_{j\in\mathcal{A}^N} p_j^N \Phi_j^N, \tag{34}$$

as the global polynomial using the approximate partitions of unity and which we will ultimately represent using an EML tree. Similarly, we define $\tilde{f}^N$ as

$$\tilde{f}^N = \sum_{j\in\mathcal{A}^N} f\,\Phi_j^N. \tag{35}$$

Step 3. Approximation of f by $\tilde{f}^N$. We first prove that $\tilde{f}^N$ is close to f. Considering an arbitrary $i \in \mathcal{A}^N$ and any $k \ge 1$, we employ the property 1, Lemmas 5 and 6, and the explicit value of α (from (22)) to get

$$\begin{aligned}\|f - \tilde{f}^N\|_{W^{k,\infty}(I_i^N)} &= \left\| f - \sum_{j\in\mathcal{A}^N} f\,\Phi_j^N \right\|_{W^{k,\infty}(I_i^N)} \\ &\le 2^k\,\|f\|_{W^{k,\infty}(I_i^N)} \left\| 1 - \sum_{v\in\mathcal{V}_d} \Phi_{i+v}^N \right\|_{W^{k,\infty}(I_i^N)} + 2^k\,\|f\|_{W^{k,\infty}(I_i^N)} \left\| \sum_{\substack{j\in\mathcal{A}^N \\ j-i\notin\mathcal{V}_d}} \Phi_j^N \right\|_{W^{k,\infty}(I_i^N)} \\ &\le 2^k\,\|f\|_{W^{k,\infty}(I_i^N)} \left(2^{kd} d\varepsilon + N^d (2k)^{2k}\alpha^k\varepsilon\right) \\ &\le 2^k\,\|f\|_{W^{k,\infty}(I_i^N)} \left(2^{kd} d + N^d (2k)^{2k} N^k (k+1)^k \max\left\{R^k, \ln^k\left(\frac{2Nk^2}{\varepsilon^{\frac{1}{k+1}} e}\right)\right\}\right)\varepsilon \\ &\le \delta(2(k+1))^{3k} \max\left\{R^k, \ln^k\left(\frac{2Nk^2}{\varepsilon^{\frac{1}{k+1}} e}\right)\right\} \frac{\mathcal{C}(d,k,s,f)}{N^{s-k}},\end{aligned} \tag{36}$$

with ε chosen to satisfy

$$\varepsilon \le \frac{\delta\mathcal{C}(d,k,s,f)}{2^{(k+1)d} d N^{s+d}\,\|f\|_{W^{k,\infty}([0,1]^d)}}. \tag{37}$$

Similarly, for $k = 0$, we have

$$\|f - \tilde{f}^N\|_{L^\infty(I_i^N)} \le \|f\|_{L^\infty(I_i^N)} (d\varepsilon + N^d\varepsilon) \le \frac{\delta}{3}\frac{\mathcal{C}(d,k,s,f)}{N^{s-k}}. \tag{38}$$

Step 4. Approximation of $\tilde{f}$ by the Global Polynomial $\tilde{p}^N$. For $k = 0$,

$$\begin{aligned}
\|\tilde{f}^N - \tilde{p}^N\|_{L^\infty(I_i^N)} &= \left\| \sum_{j\in\mathcal{A}^N} (f - p_j^N)\, \Phi_j^N \right\|_{L^\infty(I_i^N)} \\
&= \left\| \sum_{v\in\mathcal{V}_d} (f - p_{i+v}^N)\, \Phi_{i+v}^N \right\|_{L^\infty(I_i^N)} + \sum_{\substack{j\in\mathcal{A}^N \\ j-i\notin\mathcal{V}_d}} \|f - p_j^N\|_{L^\infty(I_i^N)} \|\Phi_j^N\|_{L^\infty(I_i^N)} \\
&\le \left(\max_{v\in\mathcal{V}_d} \|f - p_{i+v}^N\|_{L^\infty(I_i^N)} \right) \left\| \sum_{v\in\mathcal{V}_d} \Phi_{i+v}^N \right\|_{L^\infty(I_i^N)} + \frac{\mathcal{C}(d,k,s,f)}{N^{s-k}} N^d \varepsilon \\
&\le \frac{\mathcal{C}(d,k,s,f)}{N^{s-k}} (1 + d\varepsilon) + \frac{\mathcal{C}(d,k,s,f)}{N^{s-k}} N^d \varepsilon \\
&\le \left(1 + \frac{\delta}{3}\right) \frac{\mathcal{C}(d,k,s,f)}{N^{s-k}},
\end{aligned} \tag{39}$$

with suitably chosen ε.

Whereas for $k \ge 1$,

$$\begin{aligned}
\|\tilde{f}^N - \tilde{p}^N\|_{W^{k,\infty}(I_i^N)} &= \left\| \sum_{j\in\mathcal{A}^N} (f - p_j^N)\, \Phi_j^N \right\|_{W^{k,\infty}(I_i^N)} \\
&= \left\| \sum_{v\in\mathcal{V}_d} (f - p_{i+v}^N)\, \Phi_{i+v}^N \right\|_{W^{k,\infty}(I_i^N)} + \sum_{\substack{j\in\mathcal{A}^N \\ j-i\notin\mathcal{V}_d}} 2^k \|f - p_j^N\|_{L^\infty(I_i^N)} \|\Phi_j^N\|_{W^{k,\infty}(I_i^N)} \\
&\le \text{something } + N^d 2^k
\end{aligned} \tag{40}$$

Step 5. Finding the EML Tree. In this step, our objective is to construct an EML tree T^N such that it is exactly s exactly equal to the approximate global polynomial, that is $T^N = \tilde{p}^N$.

Step 6. Conclusion. Finally,

$$\begin{aligned}
&\|f - T^N\|_{W^{k,\infty}([0,1]^d)} \\
&\quad\le \|f - \tilde{f}^N\|_{W^{k,\infty}([0,1]^d)} + \|\tilde{f}^N - \tilde{p}^N\|_{W^{k,\infty}([0,1]^d)} + \|\tilde{p}^N - T^N\|_{W^{k,\infty}([0,1]^d)} \\
&\quad\le \|f - \tilde{f}^N\|_{W^{k,\infty}([0,1]^d)} + \|\tilde{f}^N - \tilde{p}^N\|_{W^{k,\infty}([0,1]^d)} \\
&\quad\le
\end{aligned} \tag{41}$$

Step 7. Upper Bound on the Size of the EML Tree.

□

References

Tim De Ryck, Samuel Lanthaler, and Siddhartha Mishra. On the approximation of functions by tanh neural networks. *Neural Networks*, 143:732–750, 2021.

Ricardo G Durán. On polynomial approximation in sobolev spaces. *SIAM journal on numerical analysis*, 20(5):985–988, 1983.

Andrzej Odrzywołek. All elementary functions from a single binary operator, 2026. URL `https://arxiv.org/abs/2603.21852`.

Rüdiger Verfürth. A note on polynomial approximation in sobolev spaces. *ESAIM: Mathematical Modelling and Numerical Analysis*, 33(4):715–719, 1999.

Philippe Martin Wyder, Judah Goldfeder, Alexey Yermakov, Yue Zhao, Stefano Riva, Jan P Williams, David Zoro, Amy Sara Rude, Matteo Tomasetto, Joe Germany, et al. Common task framework for a critical evaluation of scientific machine learning algorithms. *arXiv preprint arXiv:2510.23166*, 2025.

Alexey Yermakov, Yue Zhao, Marine Denolle, Yiyu Ni, Philippe M Wyder, Judah Goldfeder, Stefano Riva, Jan Williams, David Zoro, Amy Sara Rude, et al. The seismic wavefield common task framework. *arXiv preprint arXiv:2512.19927*, 2025.

NeurIPS Paper Checklist

The checklist is designed to encourage best practices for responsible machine learning research, addressing issues of reproducibility, transparency, research ethics, and societal impact. Do not remove the checklist: **The papers not including the checklist will be desk rejected.** The checklist should follow the references and follow the (optional) supplemental material. The checklist does NOT count towards the page limit.

Please read the checklist guidelines carefully for information on how to answer these questions. For each question in the checklist:

- You should answer [Yes], [No], or [N/A].
- [N/A] means either that the question is Not Applicable for that particular paper or the relevant information is Not Available.
- Please provide a short (1–2 sentence) justification right after your answer (even for [N/A]).

The checklist answers are an integral part of your paper submission. They are visible to the reviewers, area chairs, senior area chairs, and ethics reviewers. You will also be asked to include it (after eventual revisions) with the final version of your paper, and its final version will be published with the paper.

The reviewers of your paper will be asked to use the checklist as one of the factors in their evaluation. While [Yes] is generally preferable to [No], it is perfectly acceptable to answer [No] provided a proper justification is given (e.g., error bars are not reported because it would be too computationally expensive" or "we were unable to find the license for the dataset we used"). In general, answering [No] or [N/A] is not grounds for rejection. While the questions are phrased in a binary way, we acknowledge that the true answer is often more nuanced, so please just use your best judgment and write a justification to elaborate. All supporting evidence can appear either in the main paper or the supplemental material, provided in appendix. If you answer [Yes] to a question, in the justification please point to the section(s) where related material for the question can be found.

IMPORTANT, please:

- **Delete this instruction block, but keep the section heading "NeurIPS Paper Checklist"**,
- **Keep the checklist subsection headings, questions/answers and guidelines below.**
- **Do not modify the questions and only use the provided macros for your answers**.

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: **[TODO]**

   Justification: **[TODO]**

   Guidelines:

   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
   - It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: **[TODO]**

   Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate "Limitations" section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: **[TODO]**

   Justification: **[TODO]**

   Guidelines:

   - The answer [N/A] means that the paper does not include theoretical results.
   - All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
   - All assumptions should be clearly stated or referenced in the statement of any theorems.
   - The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
   - Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
   - Theorems and Lemmas that the proof relies upon should be properly referenced.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

   Answer: **[TODO]**

   Justification: **[TODO]**

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.

- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general. releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
  (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
  (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
  (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
  (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: **[TODO]**

   Justification: **[TODO]**

   Guidelines:

   - The answer [N/A] means that paper does not include experiments requiring code.
   - Please see the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
   - While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
   - The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
   - The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
   - The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
   - At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).

- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

   Answer: **[TODO]**

   Justification: **[TODO]**

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
   - The full details can be provided either with the code, in appendix, or as supplemental material.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: **[TODO]**

   Justification: **[TODO]**

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
   - The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
   - The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
   - The assumptions made should be given (e.g., Normally distributed errors).
   - It should be clear whether the error bar is the standard deviation or the standard error of the mean.
   - It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
   - For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
   - If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: **[TODO]**

   Justification: **[TODO]**

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
   - The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.

- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. **Code of ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

   Answer: **[TODO]**

   Justification: **[TODO]**

   Guidelines:

   - The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
   - If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
   - The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. **Broader impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

    Answer: **[TODO]**

    Justification: **[TODO]**

    Guidelines:

    - The answer [N/A] means that there is no societal impact of the work performed.
    - If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
    - Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
    - The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
    - The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
    - If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. **Safeguards**

    Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

    Answer: **[TODO]**

    Justification: **[TODO]**

    Guidelines:

    - The answer [N/A] means that the paper poses no such risks.

- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. **Licenses for existing assets**

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, `paperswithcode.com/datasets` has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. **New assets**

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. **Crowdsourcing and research with human subjects**

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

    Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

    Answer: **[TODO]**

    Justification: **[TODO]**

    Guidelines:

    - The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
    - Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
    - We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
    - For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. **Declaration of LLM usage**

    Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

    Answer: **[TODO]**

    Justification: **[TODO]**

    Guidelines:

    - The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
    - Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.